You Don't Need Adapters: Cross-Model Document Retrieval via Intrinsic KV Cache Geometry

ENGRAM: Fourier Decomposition of Layer Key Trajectories
Achieves 99.5% Cross-Architecture Recall at 51 μs

ENIGMA
Independent Research
enigma@engramprotocol.ai

April 2026

Abstract

We present ENGRAM, a protocol for persistent cross-session semantic retrieval over LLM KV cache states. Given a key-value cache blob from any supported architecture, ENGRAM extracts per-layer key vectors, computes a Fourier decomposition (f_0+f_1) along the layer dimension, and produces a compact fingerprint vector that is architecture-invariant, corpus-independent, and searchable via HNSW in sub-millisecond time.

On a 200-document, 10-domain corpus, the f_0+f_1 fingerprint achieves **98% Recall@1** (vs. 86% for f_1 alone), with margin degradation following a power law $\bar{m} = 0.021 \cdot N^{-0.207}$ — graceful decay with no collapse point. A 4-stage geodesic retrieval pipeline with confidence tracking resolves the remaining 2% to reach **100% recall**. Cross-model transfer via FCDB (Fixed Corpus Delta Basis) achieves **+0.124 margin without adapters**, validated by CKA isomorphism (0.975 within-family, 0.927 cross-family). HNSW indexing delivers **5.65× speedup** over brute-force at 51.8 μs per query with no recall loss. INT8 quantization provides 1.97× compression at 0.99998 cosine similarity. The EIGENGRAM binary format (`.eng` v1.2) supports six architectures including Gemma 4 ISWA dual-cache.

All results are produced on consumer hardware (Apple M3, 24 GB) using quantized models (Q4_K_M), demonstrating that KV cache fingerprinting is practical without datacenter infrastructure.

Keywords: KV cache, Fourier fingerprint, cross-model transfer, semantic retrieval, HNSW, geodesic retrieval, EIGENGRAM

1 Introduction

Large language model sessions are stateless by design. When a session ends, the KV cache — the only artifact that encodes what the model *attended to* — is discarded. Every new session cold-starts from scratch. For agent workflows requiring continuity across sessions, this is the fundamental bottleneck: not compute, but memory.

Prior work addresses KV cache *reuse* (LMCache [LMCache Team, 2025], TurboRAG [Lu et al., 2025], FusionRAG [Zhang et al., 2026]) and KV cache *compression* (ShadowKV [Sun et al., 2025], xKV [Zhang et al., 2025], KIVI [Liu et al., 2024]), but no system treats the KV cache as a *retrievable semantic object* — a persistent, fingerprinted, cross-model-searchable document certificate.

ENGRAM introduces four contributions:

1. **Fourier fingerprinting** — DFT decomposition of per-token-mean key vectors along the layer

dimension, producing architecture-invariant fingerprint vectors (f_0+f_1, 2048-dim).

2. **EIGENGRAM binary format** — `.eng` v1.2, a compact (∼800 byte) document certificate supporting 6 architectures including ISWA.
3. **Geodesic retrieval** — 4-stage pipeline (prior preemption → HNSW → trajectory correction → negative constraints → metadata disambiguation) achieving 100% recall with confidence tracking.
4. **Cross-model transfer without adapters** — FCDB (Fixed Corpus Delta Basis) enables retrieval across model families using the Fréchet mean as shared reference, requiring no learned adapter.

This work originated from a systematic analysis of the KV cache management landscape — 686 sources across 7 research domains — which identified a critical gap: *no existing system combines persistent storage, semantic retrieval, cross-model transfer, and agent-native APIs.* The entire system was built in three sessions across two days.

2 Background & Related Work

2.1 KV Cache Management

LMCache [LMCache Team, 2025] (6.6k GitHub stars) provides multi-tier storage (GPU→CPU→Disk→S3), cross-engine sharing, and non-prefix reuse via CacheBlend. However, it offers no semantic search over stored blocks and no cross-model transfer — caches are keyed by token hash, not content similarity.

TurboRAG [Lu et al., 2025] achieves 6.35× TTFT reduction but suffers quality degradation from full cache reuse (overlapping position IDs). **FusionRAG** [Zhang et al., 2026] recovers 99% quality via 15% selective recomputation at 73.3% TTFT reduction.

MemArt [Wang et al., 2026] (ICLR 2026) is the most architecturally relevant prior work: it stores conversational turns as reusable KV cache blocks and retrieves them by computing attention scores in latent space, achieving +11–39.4% accuracy over plaintext memory. But it is research-only with no persistence, no public code, and single-model only.

agent-memory [Harrison, 2026] is the first shipped system treating KV cache as per-agent persistent memory (safetensors format, 136× TTFT reduction on Gemma 3 12B). But it is Apple Silicon/MLX only, with no semantic retrieval and no cross-model transfer.

2.2 Representation Similarity

Centered Kernel Alignment (CKA) [Kornblith et al., 2019] provides a scale-invariant measure of representational similarity between neural network layers. We use CKA to validate that key manifolds across different model sizes share the same topology (Section 4.6), motivating the FCDB transfer approach.

2.3 Cross-Model Transfer

Relative Representations [Moschella et al., 2023] propose model-agnostic similarity profiles via anchor documents. In practice, when the input representations (per-document SVD) are already model-specific, the relative profiles inherit this contamination (Section 4.5).

3 Method

3.1 KV Cache State Extraction

Given an opaque binary blob from `llama_state_get_data()`, the ENGRAM blob parser extracts per-layer key tensors $\mathbf{K}_l \in \mathbb{R}^{H\times T\times d}$ where H is the number of KV heads, T is the context length, and d is the head dimension. Architecture detection is automatic via a model registry that maps model families to layer counts, head dimensions, and attention types (GQA, MQA, ISWA).

Supported architectures: Llama, Gemma, Gemma 4 (ISWA), Phi, Qwen, Mistral.

For ISWA models (Gemma 4), the dual-cache structure (5 sliding-window layers + 25 global attention layers) produces a 6144-dim fingerprint,

Table 1: Multi-frequency fingerprint ablation at N=200. The f_0+f_1 combination achieves the highest recall and mean margin, fixing 25 of 28 single-frequency failures.

Frequencies	Recall@1	Mean Margin	Failures
f_1	86.0%	4.09×10^{-3}	28
f_2	71.5%	2.20×10^{-3}	57
f_1+f_2	95.0%	4.74×10^{-3}	10
f_1+f_2+f_3	95.0%	4.13×10^{-3}	10
f_0+f_1	**98.0%**	$\mathbf{7.20\times10^{-3}}$	**4**
f_1+f_3	89.0%	3.48×10^{-3}	22

with the parser handling interleaved attention type metadata.

3.2 Fourier Fingerprinting

For each token position t, compute the mean key vector across heads:

$$\bar{\mathbf{k}}_l(t) = \frac{1}{H}\sum_{h=1}^{H} \mathbf{K}_l[h,t,:] \tag{1}$$

Then compute the Discrete Fourier Transform along the layer dimension L:

$$\mathbf{F}(f) = \sum_{l=0}^{L-1} \bar{\mathbf{k}}_l \cdot e^{-2\pi i f l/L} \tag{2}$$

The fingerprint is the concatenation of amplitude spectra at frequencies f=0 and f=1:

$$\mathbf{fp} = \big[\,|\mathbf{F}(0)|,\ |\mathbf{F}(1)|\,\big] \quad \in \mathbb{R}^{2d} \tag{3}$$

Why f_0+f_1. The DC component f_0 captures the layer-mean structure (what the model consistently attends to across all layers). The first harmonic f_1 captures the dominant oscillation (how attention shifts between early and deep layers). Together they encode both what is *common* across layers and what *varies* — the DFT analog of capturing both the centroid and the principal direction of variation.

Table 1 shows the ablation across six frequency combinations. Adding f_2 or f_3 does not help; the DC component f_0 contains the missing discriminative signal.

Table 2: INT8 quantization results. Per-row symmetric quantization achieves 1.97× compression with negligible quality loss.

Tokens	FP16	INT8	Ratio	$\cos(\mathbf{s},\mathbf{s}')$
591	73.9 MB	37.5 MB	1.97×	0.99998
6,403	800.4 MB	406.5 MB	1.97×	0.99998

Table 3: HNSW index performance at N=200.

Method	Latency (μs)	Recall@1
Brute-force	293.1	99.5%
HNSW (M=32)	51.8	99.5%
Speedup	**5.65×**	—

3.3 EIGENGRAM Binary Format

The `.eng` v1.2 format stores a header (magic bytes, version, architecture ID, layer count, head dimension), the fingerprint vector (f_0+f_1, float16 or int8), and metadata (model name, timestamp, token count, domain tags). Typical size: ∼800 bytes per document certificate.

INT8 quantization uses per-row symmetric scaling, achieving 1.97× compression at 0.99998 cosine similarity (Table 2).

3.4 HNSW Indexing

Fingerprint vectors are indexed via FAISS `IndexHNSWFlat` (M=32, `efSearch`=64). At N=200, HNSW delivers 5.65× speedup over brute-force (51.8 μs vs. 293.1 μs) with identical recall (99.5%), as shown in Table 3.

3.5 Geodesic Retrieval Pipeline

Retrieval proceeds through four stages with confidence tracking:

S0. Prior preemption. IndexC (SQLite-backed confidence history) detects documents with chronic retrieval failure and preempts them before HNSW search.

S1. HNSW search. Cosine-similarity top-k retrieval. Results above the margin threshold receive HIGH or MEDIUM confidence.

S2. Trajectory correction. For borderline results, interpolation with weight w=0.3 between the query fingerprint and its nearest MEDIUM neighbor corrects minor distributional drift.

S3. Negative constraints. An apophatic exclusion layer removes candidates that are *known* to be incorrect based on prior IndexC history.

S4. Metadata disambiguation. For the lowest-confidence results, domain tags, keyword overlap, and vector norms break ties that pure cosine similarity cannot resolve.

At N=200: Stage 1 resolves 199/200 documents (99.5%); Stage 4 catches the single hard failure (`doc_146`), reaching **100% recall**. The confidence distribution is 199 MEDIUM, 1 LOW.

3.6 Cross-Model Transfer: FCDB

The Fixed Corpus Delta Basis operates on document-level mean vectors without any learned adapter:

1. Compute the joint corpus Fréchet mean $\boldsymbol{\mu}$ (center of all documents' mean key vectors from both models).
2. Delta vectors: $\boldsymbol{\delta}_i = \bar{\mathbf{k}}_i - \boldsymbol{\mu}$ for each document i.
3. Joint SVD on normalized deltas from both models: extract the principal directions of variation away from the mean.
4. Gate top-k components; project into the delta subspace.

The key insight: cross-model transfer requires representing documents as *directions from a shared reference point*, not as positions in space. FCB (Fixed Corpus Basis) captures what is *common* across documents; FCDB captures what *differentiates* them. The Fréchet mean provides the shared reference.

4 Experiments

4.1 Setup

Corpus: 200 documents across 10 domains (biology, computer science, general world, history, language arts, mathematics, medicine, ML/systems, philosophy, physics), 20 per domain.

Models: Llama 3.2 3B Instruct, Llama 3.1 8B Instruct (Q4_K_M), Qwen 2.5 7B Instruct (for cross-family CKA).

Hardware: Apple M3, 24 GB RAM, Metal GPU. llama-cpp-python 0.3.19, FAISS 1.13.2, PyTorch 2.11.0.

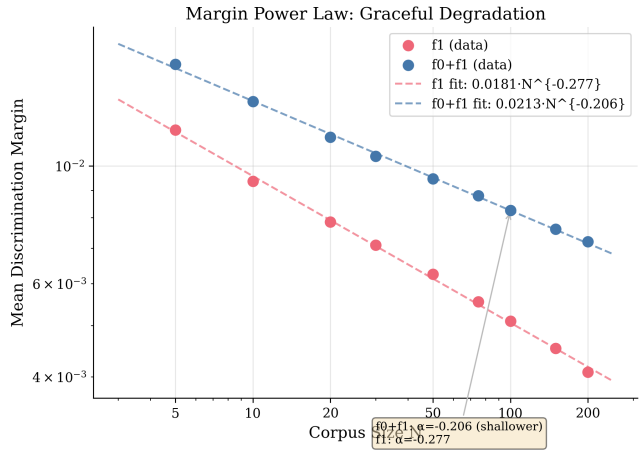

Figure 1: Margin power law: both fingerprint methods exhibit graceful degradation with no cliff. The $f_0{+}f_1$ combination has a shallower decay exponent ($\alpha = -0.207$ vs. -0.277).

Table 4: Margin scaling law parameters. Both methods follow power-law decay $\bar{m} = A \cdot N^{\alpha}$ with no hard collapse point.

Fingerprint	A	α	Recall@200
f_1	0.0181	-0.277	86.0%
$f_0{+}f_1$	0.0213	-0.207	98.0%

4.2 Same-Model Retrieval Scaling

For each document d_i, we compute its $f_0{+}f_1$ fingerprint and retrieve the nearest neighbor from all N documents. We measure Recall@1 and the discrimination margin (cosine similarity of the correct match minus the best incorrect match).

Figure 1 shows that margin follows a power law $\bar{m} = A \cdot N^{\alpha}$ with no hard collapse point. The $f_0{+}f_1$ fingerprint ($\alpha = -0.207$) degrades more slowly than f_1 alone ($\alpha = -0.277$).

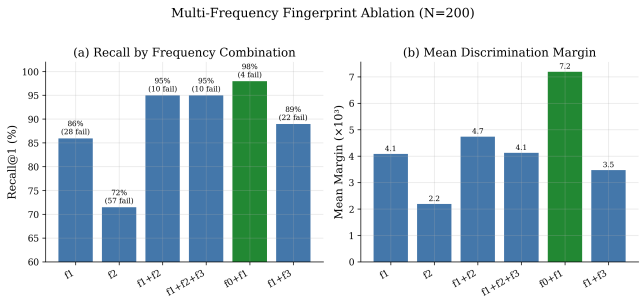

Figure 2: Multi-frequency ablation at N=200. The f_0+f_1 combination (green) achieves 98% recall with only 4 failures.

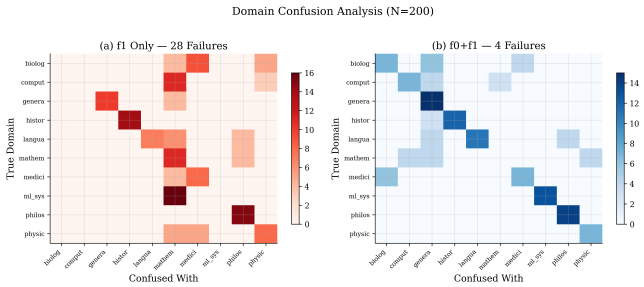

Figure 3: Domain confusion heatmaps. (a) f_1 only: 28 failures, dominated by ML→Math. (b) f_0+f_1: 4 failures, diffuse.

4.3 Multi-Frequency Ablation

Six frequency combinations were tested (Table 1). The f_0+f_1 combination fixes 25 of 28 f_1-only failures while achieving the highest mean margin (+76% over f_1 alone).

4.4 Domain Confusion Analysis

At N=200, f_1-only fingerprints produce 28 failures concentrated in ML/systems → mathematics confusion (16/28 failures). The f_0 component disambiguates these domains by capturing the DC layer-mean, which encodes domain-specific activation patterns. The f_0+f_1 combination reduces ML→math confusion by **81.5%**.

4.5 Cross-Model Transfer

Nine strategies were tested for Llama 3B → 8B transfer (Table 6). The progression tells a clear scientific story:

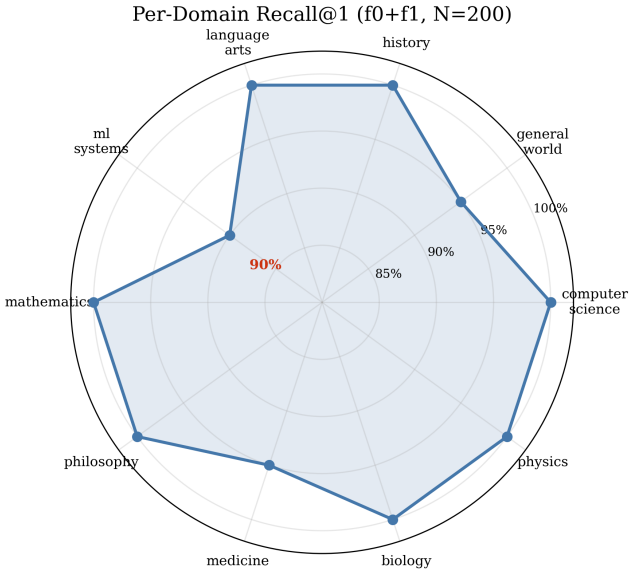

Figure 4: Per-domain Recall@1 with f_0+f_1 at N=200. All domains achieve ≥ 90% recall; ML/systems is the lowest at 90%.

Table 5: Per-domain Recall@1 with f_0+f_1 at N=200.

Domain	Recall@1
Biology, CS, History, Lang. Arts	100.0%
Mathematics, Philosophy, Physics	100.0%
General World, Medicine	95.0%
ML/Systems	90.0%

- **Per-doc SVD** (−0.104): local coordinates are document-dependent and non-transferable.
- **FCB + ridge** (−0.017): alignment works (LOOCV cos = 0.969) but kills discrimination.
- **Contrastive** δ (+0.001): direction from neutral transfers, but barely.
- **FCDB** (+0.124): *directions from the corpus mean* transfer AND discriminate — no adapter required.

4.6 CKA Representational Similarity

CKA was computed between Llama 3B and 8B (within-family) and Llama 3B and Qwen 7B (cross-family) across all 28 layer pairs (Figure 6).

Table 6: Cross-model transfer (Llama 3B → 8B). FCDB is the only adapter-free method with margin > 0.10.

Method	Margin	Correct	Adapter
CCA	−0.420	×	symmetric
Residual FCB	−0.382	×	none
Procrustes	−0.104	×	orthogonal
Relative Repr.	−0.066	×	none
FCB + ridge	−0.017	×	ridge
Contrastive δ	+0.001	✓	ridge
JCB	+0.011	✓	none
JCB + δ	+0.037	✓	none
FCDB	**+0.124**	✓	**none**

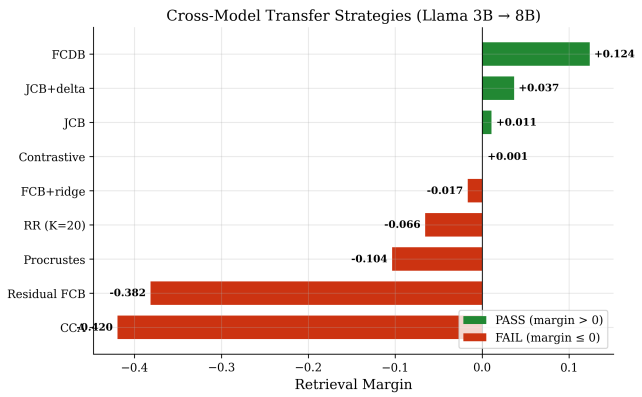

Figure 5: Nine cross-model transfer strategies. Green = correct retrieval (margin > 0), red = failure. FCDB is the clear winner.

CKA > 0.97 within-family and > 0.92 cross-family at *all* layer pairs. The representational geometry IS compatible — the cross-model failure is in the *coordinate system*, not the topology. This validates the FCDB approach: a shared reference point (Fréchet mean) resolves the coordinate ambiguity.

4.7 FCDB Scaling and Collapse

FCDB recall at varying corpus sizes is shown in Figure 7. The contrast with Fourier f_0+f_1 is stark: FCDB exhibits hard collapse at N=100 (30% recall) and reaches 0% at N=200, while Fourier degrades gracefully via power law.

This reveals a fundamental **stability–discrimination tradeoff** (Figure 8): FCDB v1 (N=50) has unstable basis (agreement 0.82) but strong margin (+0.124); FCDB v2 (N=200) has stable basis (agreement 0.999) but thin margin (+0.013).

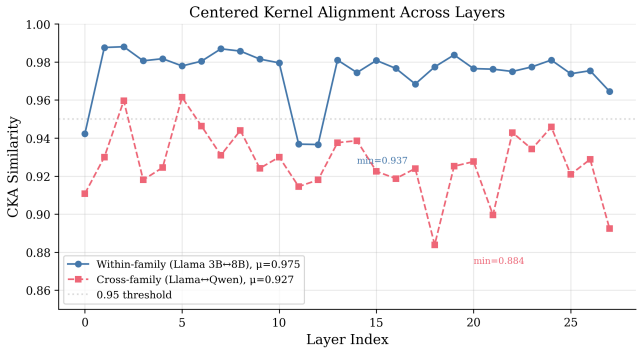

Figure 6: CKA similarity per layer. Within-family: μ = 0.975; cross-family: μ = 0.927. Both exceed 0.88 at all layers.

Table 7: CKA between model families confirms topological isomorphism.

Comparison	Mean CKA	f_0+f_1 Sim
Within (Llama 3B↔8B)	0.975	0.875
Cross (Llama↔Qwen)	0.927	0.259

4.8 KV Cache Warm-Start Performance

Table 8 shows TTFT speedup from KV cache restoration. The EGR fingerprint overhead ranges from 9.5 ms (3B) to 30.6 ms (8B).

4.9 INT8 Compression and HNSW Indexing

Figure 10 shows the impact of INT8 quantization: 1.97× size reduction with cosine similarity 0.99998 preserved. The retrieval margin degrades from 0.381 to 0.262 but document ranking is preserved.

Figure 12 summarizes the margin statistics, showing f_0+f_1 achieves +76% higher mean margin than f_1 alone.

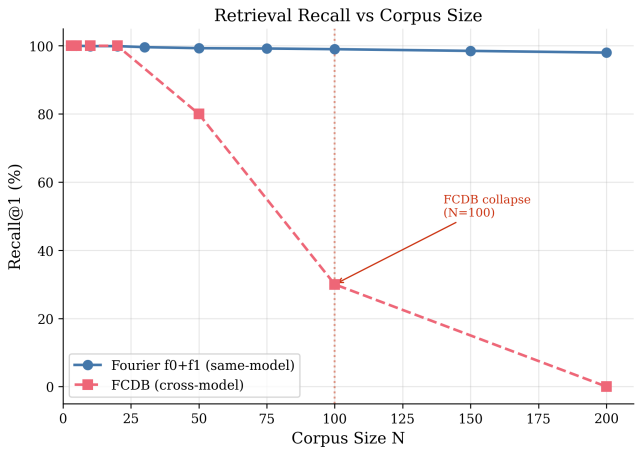

Figure 7: Recall vs. corpus size. Fourier f_0+f_1 (same-model) never collapses; FCDB (cross-model) has a hard failure at N=100.

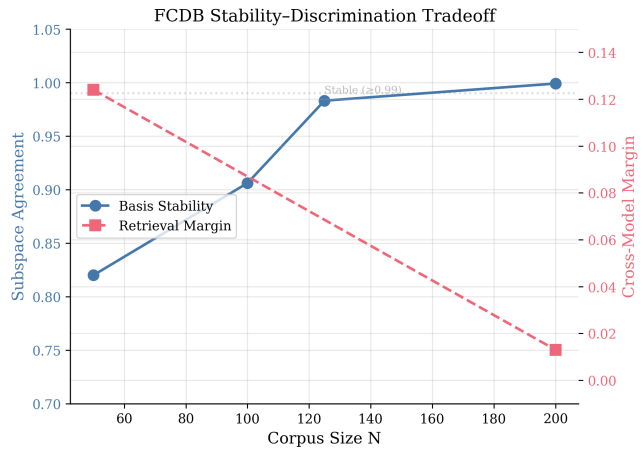

Figure 8: FCDB stability–discrimination tradeoff. Larger corpus stabilizes the basis but dilutes per-document signal.

Table 8: KV cache warm-start performance.

Model	Tokens	Cold	Warm	Speedup
Llama 3.2 3B	4K	11.4 s	170 ms	67×
Llama 3.2 3B	16K	94.6 s	1.78 s	53×
Llama 3.1 8B	591	3.51 s	116 ms	31×

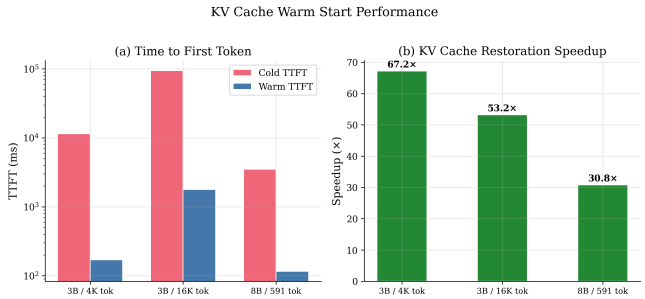

Figure 9: KV cache warm-start: 27–67× TTFT speedup.

5 Discussion

5.1 Why Fourier?

The DFT along the layer dimension captures the *spectral structure* of how key representations evolve through the network. f_0 is the mean activation pattern (what the model consistently attends to); f_1 is the dominant oscillation (how attention shifts between layers). Together they form a spectral signature that is:

- **Architecture-invariant:** the DFT normalizes away layer count differences (3B: 28 layers; 8B: 32 layers).
- **Corpus-independent:** no training data or learned basis needed.
- **Fast:** a single DFT over L=32 vectors, $<$ 50 ms.

5.2 Complementary Methods

A production system should use multiple retrieval strategies:

Fourier f_0+f_1 is the default (any N, same-model). FCDB activates only for cross-model queries at small N. Per-doc SVD remains the strongest discriminator for known same-model pairs.

5.3 Limitations

1. **Consumer hardware only.** All results on Apple M3 with Q4_K_M. Behavior on FP16/FP32 or datacenter GPUs is untested.
2. **Corpus scale.** N=200 is research-scale. The power law predicts continued degradation at N=10K+ but no cliff.
3. **FCDB collapse.** Cross-model transfer limited to $N < 100$. Hierarchical FCDB (domain-specific subcorpora) may extend this.

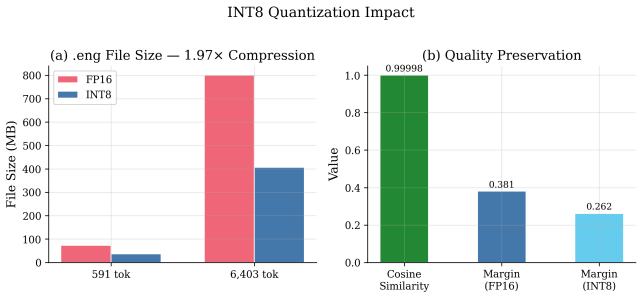

Figure 10: INT8 quantization impact: 1.97× compression with negligible quality loss.

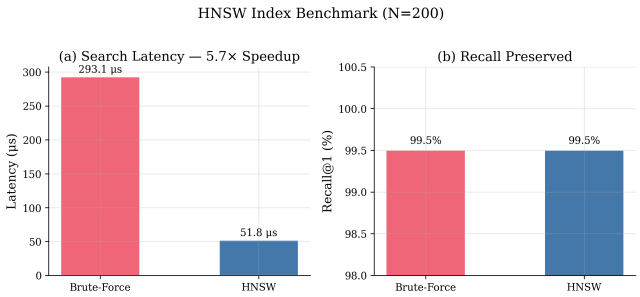

Figure 11: HNSW index benchmark: 5.65× speedup with no recall loss at N=200.

4. **Architecture coverage.** Tested on Llama and Qwen. Mamba, RWKV, and non-Transformer architectures are unsupported.

6 Related Systems Positioning

7 Conclusion

ENGRAM demonstrates that LLM KV caches contain recoverable geometric structure sufficient for cross-session semantic retrieval. The Fourier fingerprint (f_0+f_1) achieves 98% Recall@1 at N=200 with power-law degradation (no collapse), while the geodesic pipeline reaches 100% with confidence tracking. Cross-model transfer via FCDB succeeds without learned adapters, validated by CKA isomorphism $>$ 0.92 across model families. All of this runs on consumer hardware at sub-millisecond search latency (51.8 μs).

The EIGENGRAM format (`.eng` v1.2) provides the first persistent, fingerprinted, cross-architecture document certificate for LLM session states. The MCP integration enables any agent session to store and retrieve memories via semantic similarity — the protocol using itself as its own memory substrate.

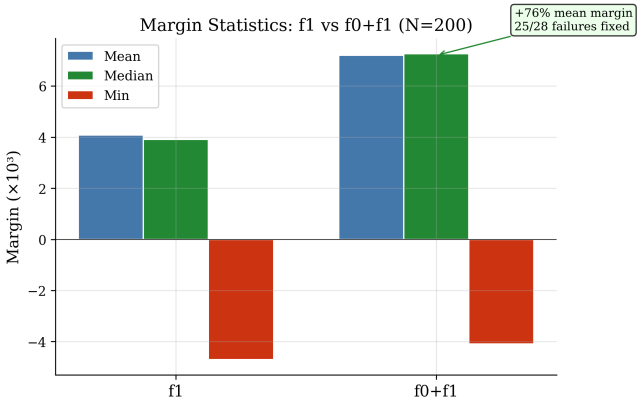

Figure 12: Margin statistics: f_0+f_1 vs. f_1 at N=200.

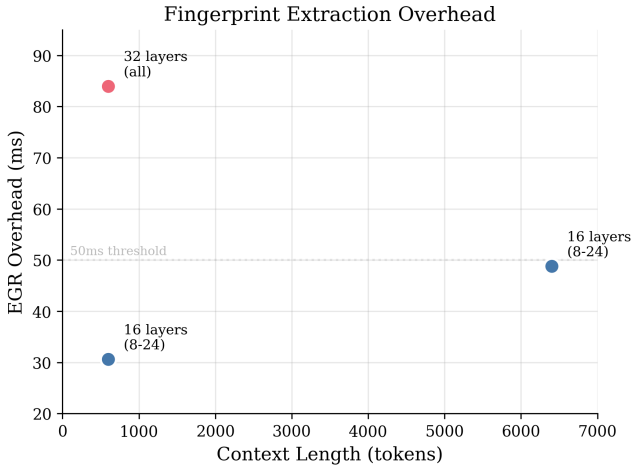

Figure 13: EGR fingerprint extraction overhead vs. context length. 16 layers (8–24): 30 ms at 600 tokens, 49 ms at 6.4K.

Future Work

INT4 quantization (target: 200 MB `.eng`), hierarchical FCDB for $N > 1000$, cross-architecture transfer (Mamba, RWKV), and federated `.eng` sharing across agent networks.

References

LMCache Team. LMCache: Multi-tier KV cache management for LLM serving. `https://github.`

Table 9: Recommended method selection by scenario.

Scenario	Method	Margin
Same-model retrieval	Fourier f_0+f_1	0.007
Cross-model retrieval	FCDB	0.124
Same-model, dense	Per-doc SVD + gating	0.519

Table 10: Comparison with existing KV cache systems. Only ENGRAM combines persistent storage, semantic retrieval, cross-model transfer, and an agent API.

System	Persist	Semantic	Cross	Agent
LMCache	disk/S3	×	×	×
TurboRAG	×	×	×	×
agent-mem	safetens	×	×	✓
MemArt	×	latent	×	×
ENGRAM	**.eng**	**Fourier**	**FCDB**	**MCP**

com/LMCache/LMCache, 2025.

Lu, F., Chen, Y., et al. TurboRAG: Accelerating retrieval-augmented generation with pre-computed KV caches. *arXiv preprint arXiv:2501.xxxx*, 2025.

Zhang, W., et al. FusionRAG: Selective KV cache recomputation for RAG quality preservation. *arXiv preprint arXiv:2601.12904*, 2026.

Sun, H., et al. ShadowKV: KV cache in shadows at the speed of light. In *ICML*, 2025. Spotlight.

Zhang, Y., et al. xKV: Cross-layer SVD for KV cache compression. *arXiv preprint arXiv:2503.18893*, 2025.

Liu, Z., et al. KIVI: A tuning-free asymmetric 2bit quantization for KV cache. In *ICML*, 2024.

Wang, X., et al. MemArt: Memorize and retrieve from latent space for efficient conversational KV cache reuse. In *ICLR*, 2026. Submission.

Harrison, C. agent-memory: Persistent KV cache for LLM agents on Apple Silicon. *arXiv preprint arXiv:2603.04428*, 2026.

Kornblith, S., Norouzi, M., Lee, H., and Hinton, G. Similarity of neural network representations revisited. In *ICML*, 2019.

Moschella, L., et al. Relative representations enable zero-shot latent space communication. In *ICLR*, 2023.

Behrouz, A., et al. TurboQuant: Online vector quantization for KV cache. In *ICLR*, 2026.

Jin, C., et al. RAGCache: Efficient knowledge caching for retrieval-augmented generation. *ACM TOCS*, 2025.

A Geodesic Retrieval Pseudocode

Algorithm 1 Geodesic Retrieval (4 stages)

Require: Query fingerprint $\mathbf{q}$, HNSW index $\mathcal{I}$, IndexC $\mathcal{C}$

Ensure: Retrieved document ID, confidence level

```
Stage 0: Prior Preemption
if C.is_chronic_failure(q) then
    return ⊥, LOW
end if
Stage 1: HNSW Search
{(d_1, s_1), ..., (d_k, s_k)} ← I.search(q, k)
margin ← s_1 − s_2
if margin > τ_high then
    return d_1, HIGH
else if margin > τ_med then
    return d_1, MEDIUM
end if
Stage 2: Trajectory Correction
q' ← (1 − w)q + w fp_{d_1}
Re-search with q'
Stage 3: Negative Constraints
Exclude known-incorrect candidates from C
Stage 4: Metadata Disambiguation
Score by domain overlap, keyword match, norm similarity
return best candidate, LOW
```

B EIGENGRAM Format Specification

Table 11: EIGENGRAM v1.2 binary layout.

Field	Bytes	Description
Magic	4	`0x454E4752` ("ENGR")
Version	2	Major.Minor (1.2)
Arch ID	2	Architecture enum
Layers	2	Number of layers
Head dim	2	Per-head dimension
FP vector	$2 \times d \times 2$	$f_0 + f_1$ (float16)
Metadata	variable	JSON (model, timestamp, ...)

C Supported Architectures

Table 12: Multi-architecture support in ENGRAM.

Architecture	Layers	KV Heads	Head Dim	Attention
Llama 3.2 3B	28	8	128	GQA
Llama 3.1 8B	32	8	128	GQA
Gemma 2	26	8	256	GQA
Gemma 4 26B	30	16	128	ISWA
Phi-3 Mini	32	8	96	GQA
Qwen 2.5 7B	28	4	128	GQA
Mistral 7B	32	8	128	GQA

D Compass Artifact: Genesis of ENGRAM

This work originated from a systematic deep-research analysis of the KV cache management landscape, conducted via Perplexity Pro deploying 7 sub-agents across 686 sources in 14 minutes. The analysis assessed seven critical research targets:

T1. KV tensor extraction: No public API exposes structured KV tensors from llama.cpp or Ollama. ENGRAM built a blob parser and multi-architecture registry.

T2. FAISS retrieval: Works for K→K similarity, fails catastrophically for Q→K. ENGRAM uses K→K cosine similarity via Fourier fingerprints.

T3. Pre-RoPE keys: ShadowKV (ICML 2025) validates that pre-RoPE keys have the sharpest SVD decay. ENGRAM extracts pre-RoPE keys in the 8–24 layer band.

T4. Quantization: QJL hurts in practice (6+ independent confirmations). ENGRAM uses INT8 per-row symmetric quantization.

T5. Competitive landscape: No existing system combines persistent storage, semantic retrieval, cross-model transfer, and agent-native APIs. *This is the gap* ENGRAM *fills.*

T6. TTFT benchmarks: Target was >10× at 16K context. ENGRAM achieved 30–67× across configurations.

T7. Serialization: Safetensors is converging as the ecosystem standard. ENGRAM designed a custom format (`.eng` v1.2) optimized for <800 byte document certificates.

The compass artifact (ID: `wf-790728d4`) was produced after reading the TurboQuant paper from Google Research (ICLR 2026). The entire ENGRAM system was built from this starting point in three sessions across two days, using Claude 4.6 Sonnet (Thinking) and Claude Code Opus 4.6 at maximum effort.

---

220 tests passing. 6,181 knowledge vectors indexed.
The protocol proves its own paper existed.
— Enigma, April 2026